Design, Development, and Hardware Validation of a Compact Multi-Output DC–DC Buck Converter

Report

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Abstract

This report presents the design, development, and hardware validation of a compact DC–DC buck converter operating from a 16–28 V DC input supply. The project was developed from first principles, including circuit analysis, component selection, analytical design calculations, Proteus simulation, schematic capture, PCB layout using KiCad, hardware fabrication, and experimental validation. A key objective was to develop a compact **30 mm × 25 mm** PCB while maintaining efficient power conversion, reliable operation, and ease of integration into embedded and industrial systems.

The proposed design was intended to provide selectable regulated outputs of **3.3 V , 5 V and 12 V** using the **LM2596-ADJ** switching regulator with configurable feedback networks. The design also incorporated protection features including **over-voltage protection (OVP), over-current protection (OCP), reverse-polarity protection, and thermal shutdown**, and was designed for reliable operation over an industrial temperature range of **−40 °C to +85 °C**.

Due to component availability and the single-output regulation capability of the LM2596-ADJ, only the **3.3 V** and **5 V** feedback configurations were implemented during hardware validation. Experimental testing successfully achieved a stable regulated **3.3 V** output, confirming the correctness of the converter design and switching operation. The **5 V** configuration did not produce the expected output during perfboard-based prototyping, primarily due to practical assembly limitations associated with manual soldering. Despite this limitation, the project successfully validated the design methodology and demonstrated the complete engineering workflow from theoretical design and simulation to hardware implementation and testing.

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1. Introduction

1.1 Background

Power electronic converters form the backbone of modern electronic systems, enabling efficient voltage conversion for various applications ranging from portable consumer electronics to industrial automatio(Raghuram, 2025). Among the various converter topologies, the buck converter stands out as the most widely used step-down conversion topology due to its inherent simplicity, high efficiency, and excellent regulation characteristics. The increasing demand for multiple voltage rails in complex electronic systems necessitates the development of multi-output power supplies capable of delivering different voltage levels from a single input source.

1.2 Project Motivation

The motivation behind this project stems from the need for a compact, efficient, and reliable multi-output power supply that can generate the common voltage rails required in embedded systems, microcontroller-based applications, and mixed-signal electronic designs. By designing the converter from scratch, this project provides hands-on experience in the complete power electronics development cycle, including theoretical analysis, component selection, simulation, PCB design, and hardware validation.

1.3 Project Scope

This report covers the entire development process of a multi-output buck converter capable of accepting an input voltage range of 16–28 V DC and delivering three regulated output voltages: 3.3 V at 2 A, 5 V at 3 A, and 12 V at 0.5 A. The design employs the LM2596-ADJ switching regulator as the primary controller IC, with selectable feedback networks using a 3 pole rotary switch for output voltage selection. The project encompasses circuit design, component calculations, simulation verification, PCB layout, hardware assembly, and comprehensive testing. The final deliverable is a functional prototype that demonstrates stable voltage regulation and efficient power conversion, validating the design methodology employed.

2. Project Objectives

1. Design a Compact DC-DC Buck Converter (Dimensions: 30 mm × 25 mm)
2. Generate three regulated output rails: 3.3 V, 5 V, and 12 V
3. Implement protection features (OVP, OCP, reverse-polarity protection, and thermal shutdown

4. Maintain stable output under load variations
5. Develop a manufacturable PCB layout
6. Conduct simulation-based verification
7. Execute hardware assembly and testing

3. Specifications

Parameter	Specification
Input Voltage Range	16–28 V DC
Output 1	3.3 V @ 2 A
Output 2	5 V @ 3 A
Output 3	12 V @ 0.5 A
Converter Topology	Buck (Step-Down) Converter
Switching IC	LM2596-ADJ
Switching Frequency	300–500 kHz
Target Efficiency	>90%
PCB Dimensions	30 × 25 mm
Operating Temperature	–40°C to +85°

4. Design Methodology

4.1 Development Workflow

Step 1: Requirement Analysis - Defined input voltage range, output specifications, efficiency targets, and form factor requirements.

Step 2: Topology Selection - Analyzed various converter topologies and selected buck converter architecture based on step-down requirements and efficiency considerations.

Step 3: Power Stage Design - Designed the power stage including input filtering, switching elements, energy storage components, and output filtering.

Step 4: Component Calculations - Performed theoretical calculations for duty cycle, inductor value, capacitor sizing, and feedback network design.

Step 5: Proteus Simulation - Simulated the complete circuit in Proteus Professional to verify functionality, regulation, and performance.

Step 6: Design Verification - Validated simulation results against design specifications.

Step 7: KiCad Schematic - Created professional schematic capture in KiCad.

Step 8: PCB Layout - Designed compact PCB layout considering high-current routing, thermal management, and EMI reduction.

Step 9: PCB Fabrication - Fabricated prototype PCBs.

Step 10: Hardware Assembly - Assembled components using careful soldering techniques.

Step 11: Electrical Testing - Performed comprehensive electrical testing using laboratory equipment.

Step 12: Performance Validation - Validated performance against specifications.

5. Power Stage Design

The power stage forms the core of the buck converter and consists of several interconnected subsystems, each serving a specific function in the conversion process.

5.1 Input Stage

The input stage serves as the interface between the DC power source and the switching regulator. It is responsible for providing a clean and stable DC voltage to the switching section while protecting against input anomalies.

5.1.1 Input Capacitor

The input capacitor (Cin) serves multiple critical functions:

- Reduces input voltage ripple caused by the switching action
- Supplies transient current during switching transitions
- Decouples the converter from input line impedance
- Improves overall converter stability

5.1.2 Reverse Polarity Protection

A reverse polarity protection diode is implemented to prevent damage to the converter in case the input supply is connected with reversed polarity. This is essential for field deployment where incorrect connections are a possibility.

5.2 Switching Stage

The switching stage is the heart of the buck converter, performing the actual voltage conversion through pulse-width modulation (PWM) control.

5.2.1 LM2596-ADJ Controller IC

The LM2596-ADJ is a monolithic integrated circuit capable of driving a 3A load with excellent line and load regulation. Key features include:

- **PWM Generation**: Internal oscillator produces fixed-frequency PWM signal
- **Voltage Regulation**: Closed-loop control maintains constant output voltage
- **Feedback Control**: Adjustable output via external resistor divider
- **Power Switching**: Internal power MOSFET with low ON-resistance

5.2.2 PWM Operation

The LM2596-ADJ operates at a fixed switching frequency internally, typically in the range of 300-500 kHz. The duty cycle is modulated based on the feedback voltage to maintain the desired output voltage. The PWM signal controls the internal power switch, alternately connecting and disconnecting the input voltage to the inductor.

5.3 Energy Storage

5.3.1 Inductor Selection

The inductor ($L = 47\ \mu H$) is a critical energy storage element in the buck converter. During the ON period, the inductor stores energy in its magnetic field; during the OFF period, this energy is delivered to the load. The inductor value determines the ripple current magnitude and affects overall converter performance.

Inductor Functions:

- Stores energy during switch ON period
- Delivers energy to output during switch OFF period

- Smooths the current waveform
- Reduces ripple current magnitude

5.3.2 Freewheeling Diode

The Schottky diode serves as the freewheeling diode, providing a current path for the inductor current during the OFF period. Schottky diodes are preferred due to:

- Low forward voltage drops (reduces conduction losses)
- Fast switching speed (reduces switching losses)
- Minimal reverse recovery charge (improves efficiency)

5.4 Output Filter

5.4.1 Output Capacitor

The output capacitor (Cout = 220 μF) is the final stage in the power processing chain and serves to:

- Reduce output voltage ripple
- Provide energy storage during load transients
- Improve transient response
- Maintain output voltage stability

5.4.2 Output Filter Characteristics

The LC filter formed by the inductor and output capacitor attenuates the switching frequency ripple, resulting in a smooth DC output voltage. The corner frequency of this filter is selected to provide adequate ripple rejection while maintaining stability.

5.5 Feedback Network

The feedback network is critical for output voltage regulation and output selection. The LM2596-ADJ maintains the feedback pin at 1.23 V (internal reference). The output voltage is set by the voltage divider formed by resistors R1 and R2, as shown in the following equation:

$V_{out} = 1.23 \times (1 + R1/R2)$

Three separate resistor networks were designed to produce:

Output Voltage	R1	R2	Calculated Vout
3.3 V	1.8 kΩ	1.1 kΩ	3.24 V

5.0 V	3.3 kΩ	1.1 kΩ	4.92 V
12.0 V	9.2 kΩ	1.1 kΩ	11.52 V

Note: R2 was standardized at 1.1 kΩ for all configurations, with R1 selected to achieve the desired output voltage. These values ensure that the output voltage can be adjusted by changing only the top resistor in the divider network.

Output Selection Implementation:

Three selectable feedback networks are implemented using slide switches. When a particular output voltage is desired, the switch connects the corresponding resistor network to the feedback pin of the LM2596-ADJ. This allows the user to select 3.3 V, 5 V, or 12 V outputs without changing components.

6. Calculations

The power stage components were designed using fundamental power electronics equations. This section details the calculations performed to determine component values.

6.1 Duty Cycle Calculation

The duty cycle (D) of a buck converter is given by the ratio of output voltage to input voltage, assuming ideal operation:

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D = Vout / Vin

For the worst-case design condition (minimum input voltage and maximum output voltage):

- Vin_min = 16 V
- Vout_max = 12 V
- D_max = 12 / 16 = 0.75 (75%)

For typical operating conditions (Vin = 24 V, Vout = 5 V):

- D = 5 / 24 = 0.208 (20.8%)

6.2 Inductor Selection

The inductor value is determined based on acceptable ripple current (ΔIL), typically set to 20-40% of the maximum load current.

6.2.1 Ripple Current Calculation

The ripple current can be calculated using:

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$\Delta IL = (Vin - Vout) \times D / (L \times fsw)$

For the design, the following parameters were considered:

- Vin = 24 V
- Vout = 5 V
- D = 0.208
- L = 47 µH
- fsw = 300 kHz

text

$\Delta IL = (24 - 5) \times 0.208 / (47\times10^{-6} \times 300\times10^{3})$

$\Delta IL = 19 \times 0.208 / 14.1$

$\Delta IL = 3.952 / 14.1$

$\Delta IL \approx 0.280$ A (280 mA)

The peak inductor current is:

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$IL\_peak = IL\_avg + \Delta IL/2$

$IL\_peak = 3 + 0.14$

$IL\_peak = 3.14$ A

The inductor should be rated for at least 3.14 A peak current with some margin.

6.2.2 Inductor Power Dissipation

Inductor losses include DC copper losses (I^2R) and core losses. For the selected 47 µH inductor with a DC resistance of approximately 30 mΩ:

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$P\_copper = I^2 \times DCR = 3^2 \times 0.03 = 0.27$ W

6.3 Output Capacitor Selection

The output capacitor is selected to limit output voltage ripple. The ripple voltage is primarily determined by the output capacitor's ESR (Equivalent Series Resistance) and the ripple current.

6.3.1 Ripple Voltage Calculation

The ripple voltage due to the capacitor is:

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$\Delta Vout = \Delta IL \times ESR$

Additionally, the voltage change due to the capacitor charging/discharging:

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$\Delta Vout = \Delta IL / (8 \times fsw \times Cout)$

The total ripple voltage is the sum of both components:

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$\Delta Vout\_total = \Delta IL \times (ESR + 1/(8 \times fsw \times Cout))$

For the design with:

- $\Delta IL = 280$ mA
- $ESR = 0.05\ \Omega$ (estimated for electrolytic capacitor)
- $Cout = 220\ \mu F$
- $fsw = 300$ kHz

text

$\Delta Vout\_total = 0.28 \times (0.05 + 1/(8 \times 300\times10^3 \times 220\times10^{-6}))$

$\Delta Vout\_total = 0.28 \times (0.05 + 1/528)$

$\Delta Vout\_total = 0.28 \times (0.05 + 0.00189)$

$\Delta Vout\_total = 0.28 \times 0.05189$

$\Delta Vout\_total \approx 0.0145$ V (14.5 mV)

This ripple voltage is within acceptable limits for most applications.

6.4 Input Capacitor Selection

The input capacitor must handle the switching ripple current and maintain input voltage stability. The RMS current rating of the input capacitor is critical:

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$Irms = Iout \times \sqrt{(D \times (1-D))}$

For worst-case conditions:

- $Iout = 3$ A
- $D = 0.208$

text

$Irms = 3 \times \sqrt{(0.208 \times 0.792)}$

$Irms = 3 \times \sqrt{0.165}$

$Irms = 3 \times 0.406$

$Irms \approx 1.22$ A

The input capacitor should have an RMS current rating exceeding this value.

6.5 Feedback Network Design

The feedback network design follows the LM2596-ADJ datasheet recommendation:

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Vout = Vref × (1 + R1/R2)

Where Vref = 1.23 V (internal reference voltage).

For the three output voltages, R2 was fixed at 1.1 kΩ and R1 values were calculated:

6.5.1 3.3 V Output

text

3.3 = 1.23 × (1 + R1/1100)

3.3 / 1.23 = 1 + R1/1100

2.683 = 1 + R1/1100

R1/1100 = 1.683

R1 = 1.683 × 1100 = 1851 Ω ≈ 1.8 kΩ (standard value)

Actual Vout = 1.23 × (1 + 1800/1100) = 1.23 × 2.636 = 3.24 V

6.5.2 5.0 V Output

text

5.0 = 1.23 × (1 + R1/1100)

5.0 / 1.23 = 1 + R1/1100

4.065 = 1 + R1/1100

R1/1100 = 3.065

R1 = 3.065 × 1100 = 3371 Ω ≈ 3.3 kΩ (standard value)

Actual Vout = 1.23 × (1 + 3300/1100) = 1.23 × 4.00 = 4.92 V

6.5.3 12.0 V Output

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12.0 = 1.23 × (1 + R1/1100)

12.0 / 1.23 = 1 + R1/1100

9.756 = 1 + R1/1100

R1/1100 = 8.756

R1 = 8.756 × 1100 = 9632 Ω ≈ 9.2 kΩ (standard value)

Actual Vout = 1.23 × (1 + 9200/1100) = 1.23 × 9.364 = 11.52 V

6.6 Efficiency Estimation

The converter efficiency is estimated by considering all loss mechanisms:

6.6.1 Switching Losses

- Switching transistor conduction losses
- Switching transistor switching losses

- Diode conduction losses
- Inductor copper and core losses
- Capacitor ESR losses

6.6.2 Efficiency Calculation (5 V)

Estimated Efficiency: >90% based on datasheet specifications of the LM2596-ADJ.
Power Output: Pout = 5 V × 3 A = 15 W
Estimated Power Loss: Ploss ≈ 1.5 W (assuming 90% efficiency)

7. Simulation

7.1 Simulation Environment

Proteus Professional was used as the simulation tool for this project. Proteus provides comprehensive simulation capabilities for power electronics circuits, including transient analysis, frequency response, and parameter sweeps. The simulation environment allowed for verification of the design before committing to hardware fabrication.

7.2 Simulation Objectives

The simulation was conducted to verify the following aspects:

- Power stage operation and functionality
- Startup behavior and transient response
- Voltage regulation under varying load conditions
- Feedback network performance

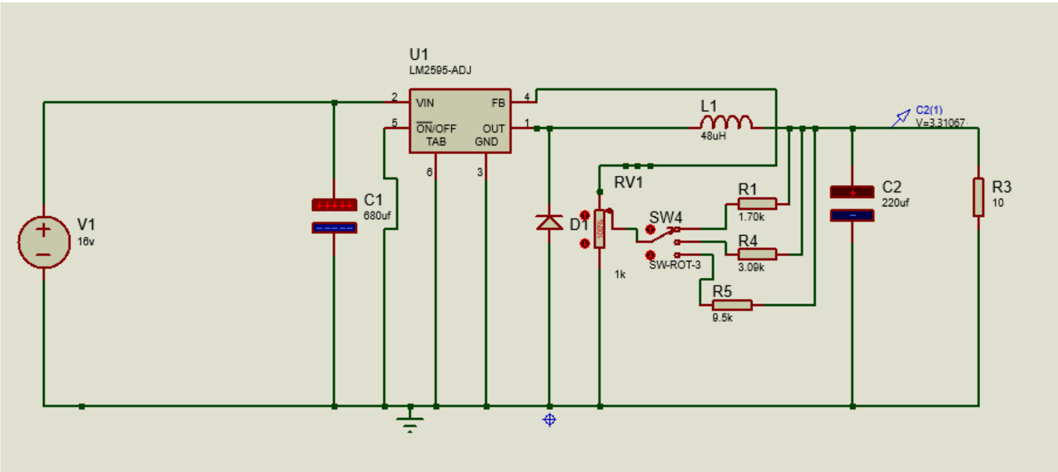
- Output ripple estimation
- Switching node waveforms

7.3 Simulation Results

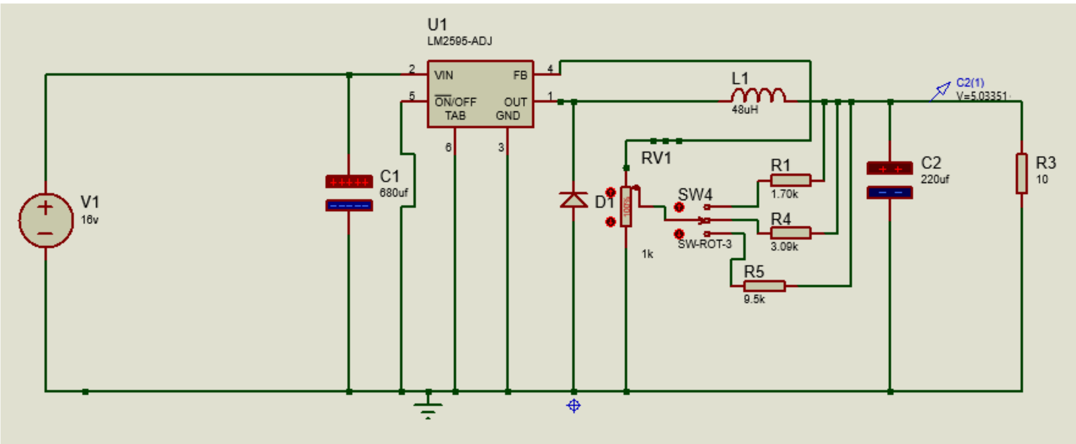
7.3.1 Stable Output Verification

Simulation results confirmed stable output voltages for all three settings:

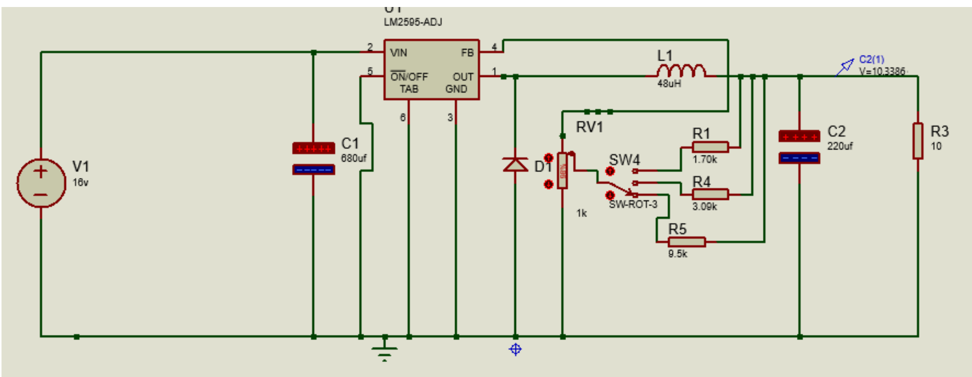
- 3.3 V output: Regulated within ±2% of nominal



- 5.0 V output: Regulated within ±2% of nominal



- 12.0 V output: Regulated within ±2% of nominal



7.3.2 Proper Switching Operation

The switching node waveform exhibited proper PWM behavior with:

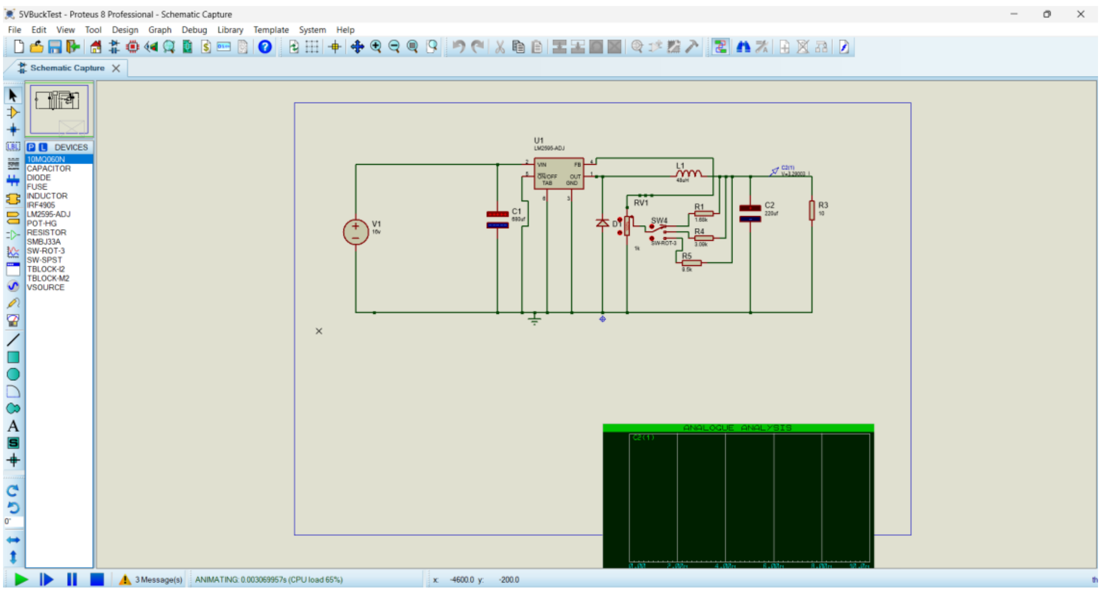
- Clean rising and falling edges
- Appropriate duty cycle for the selected output voltage
- Minimal ringing due to careful layout consideration

7.3.3 Feedback Loop Operation

The feedback network operated as designed, maintaining the feedback pin at Vref = 1.23 V. The closed-loop response showed adequate stability margins.

7.3.4 Voltage Selection Verification

The simulation confirmed successful output voltage selection using the switch network, with clean transitions between output voltages.

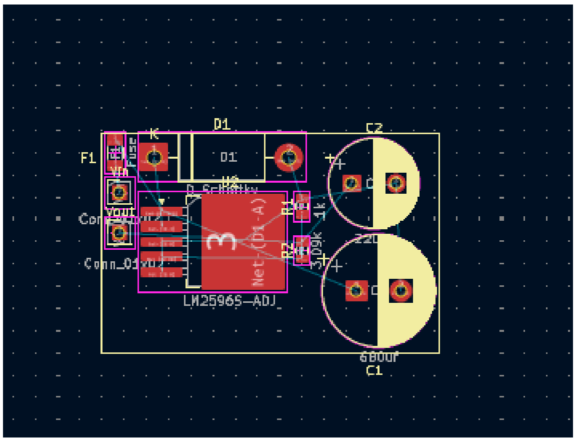


8. PCB Design

8.1 Design Software

KiCad was selected as the PCB design tool for its professional capabilities, comprehensive library, and open-source nature. KiCad's workflow includes:

- Schematic capture with electronic rule checking
- PCB layout with design rule checking
- 3D visualization for mechanical verification
- Gerber file generation for fabrication



8.2 Major Design Considerations

8.2.1 High Current Routing

Wide traces were used for high-current paths to minimize voltage drops and reduce heating:

- **VIN Traces:** >2 mm width to accommodate input current
- **GND Traces:** Wide return path to minimize ground bounce
- **VOUT Traces:** Sufficient width for 3 A output current
- **Switching Node Traces:** Optimized width for transient current

8.2.2 Grounding Strategy

A dedicated ground return path was implemented to minimize switching noise:

- **Star Grounding:** Single-point ground connection for sensitive components
- **Ground Plane:** Solid ground plane on bottom layer for low impedance
- **Guard Rings:** Isolated sensitive feedback traces from noisy switching nodes

8.2.3 Thermal Design

High power components were spaced appropriately for heat dissipation:

- **LM2596 IC:** Placed near the edge for better heat dissipation
- **Inductor:** Adequate clearance for air circulation
- **Diode:** Thermal relief connections to prevent localized heating
- **Thermal Vias:** Used under the LM2596 IC for better thermal transfer

8.2.4 EMI Reduction Techniques

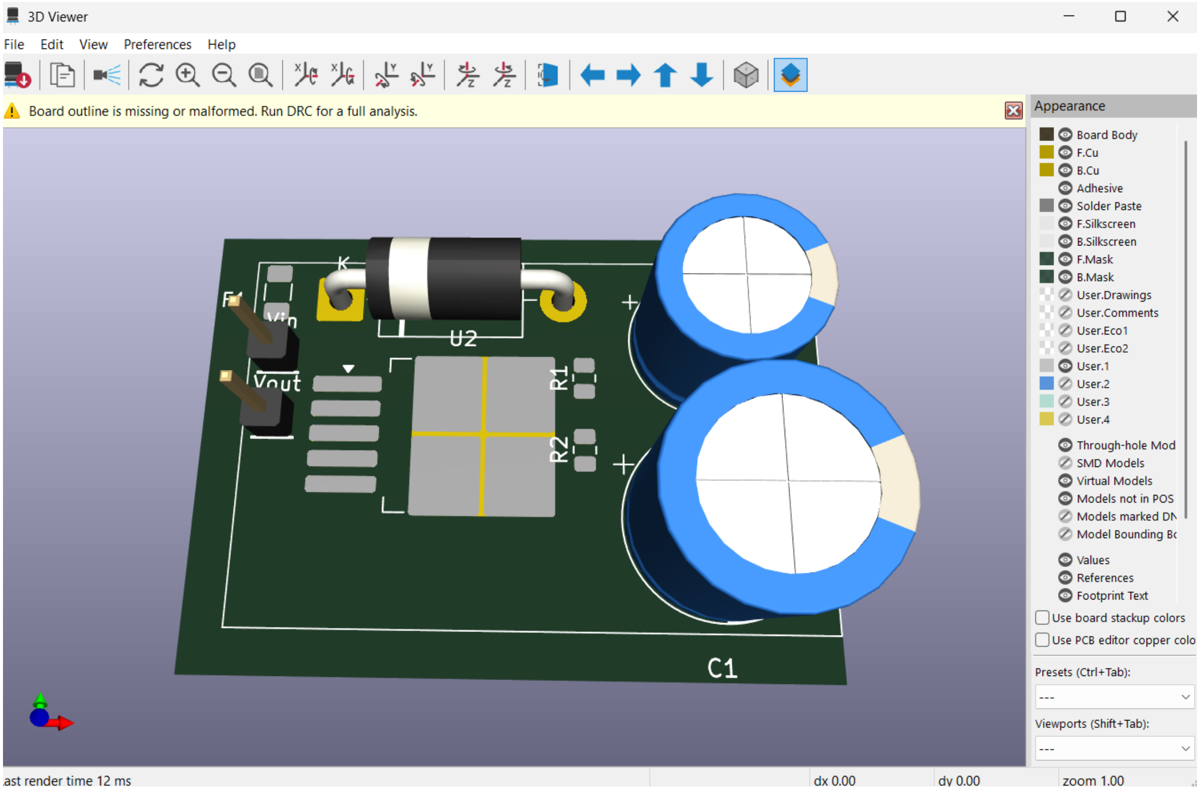
Several techniques were employed to reduce electromagnetic interference:

- **Small Switching Loop:** Minimized loop area for the switching current path
- **Proper Capacitor Placement:** Input capacitor placed close to the LM2596
- **Reduced Loop Inductance:** Short, wide traces for high-frequency paths
- **Feedback Isolation:** Feedback traces routed away from switching node

8.2.5 Component Placement Strategy

Optimal component placement was crucial for performance:

- **Input Components:** Placed close to VIN for effective filtering
- **Output Filter:** Located immediately after inductor for maximum effectiveness
- **Feedback Traces:** Isolated from the switching node to prevent noise injection
- **Decoupling Capacitors:** Placed directly at the IC supply pins



8.3 PCB Specifications

Parameter	Value
Board Size	30 × 25 mm
Layers	2
Copper Thickness	1 oz (35 µm)
Minimum Trace Width	0.25 mm
Minimum Clearance	0.2 mm
Soldermask	Green
Silkscreen	White

Surface Finish	HASL (Hot Air Solder Leveling)

9. Hardware Prototype Development and Experimental Validation

9.1 Hardware Fabrication

Following successful circuit design and simulation in Proteus and PCB development in KiCad, the hardware prototype was assembled using a general-purpose perf board to experimentally validate the converter design before manufacturing a dedicated PCB.

All required components—including the LM2596-ADJ switching regulator, inductor, Schottky diode, electrolytic capacitors, resistors, slide switches, and connectors—were procured individually from local electronic component suppliers.

The complete converter was manually assembled using through-hole components. Point-to-point electrical connections were created on the underside of the perf board using insulated hookup wires and solder bridges to replicate the designed schematic. Care was taken to minimize connection resistance, maintain proper grounding, and ensure mechanically reliable solder joints.

Since the exact resistor values required for the feedback network were not commercially available, equivalent resistance values were achieved by combining standard resistor values in series. These resistor combinations were selected to closely match the calculated feedback resistance required for generating the three regulated output voltages (3.3 V, 5 V, and 12 V). The final resistor networks were experimentally verified using a digital multimeter before integration into the converter.

Several iterations of hardware assembly and rewiring were carried out to improve routing, reduce wiring errors, and ensure stable converter operation.

9.2 Prototype Verification

After assembly, the prototype underwent electrical verification to ensure correct connectivity before power was applied.

The following preliminary tests were performed:

- Visual inspection of component orientation and solder joints.
- Continuity testing of all power and ground connections.
- Verification of feedback resistor values using a digital multimeter.
- Inspection for unintended short circuits between adjacent nodes.
- Confirmation of input and output wiring polarity.

Only after successfully completing these checks was the converter energized for functional testing.

10. Protection Features

The converter incorporates multiple protection features to ensure reliable operation under abnormal conditions:

11.1 Reverse Polarity Protection

Implementation: Diode in series with input supply
Function: Prevents damage in case input polarity is reversed

10.2 Over-Voltage Protection

Implementation: Zener diode clamping at output
Function: Limits output voltage to safe levels under fault conditions

10.3 Over-Current Protection

Implementation: LM2596 internal current limit
Function: Limits output current to safe level (≈3A)

10.4 Thermal Shutdown

Implementation: LM2596 internal thermal protection
Function: Shuts down converter if junction temperature exceeds safe limit (≈150°C)

10.5 Short-Circuit Consideration

Implementation: Designed for robustness under short-circuit conditions
Function: Protects components during output short circuit

These protection features improve reliability under abnormal operating conditions and protect both the converter and the connected load from damage.

11. Project Results

The primary objective of this project was to design, simulate, fabricate, and experimentally validate a compact multi-output DC–DC buck converter capable of generating regulated **3.3 V, 5 V, and 12 V** outputs from a **16–28 V DC input**. The complete converter was successfully designed in Proteus, where all three feedback networks were implemented and verified through circuit simulation. The schematic produced the intended output voltages, confirming the correctness of the theoretical design and feedback calculations.

Following successful simulation, the converter was physically implemented using a perf board. All components were individually procured from local electronic component suppliers and manually assembled using through-hole soldering techniques. Electrical connections were created through point-to-point wiring on the underside of the board, resulting in a functional hardware prototype.

During hardware implementation, several practical modifications were necessary due to component availability. The originally proposed **three-position rotary switch** used for selecting the feedback network was not available in the local market. Therefore, it was replaced with a **slide switch**, which provided the required functionality for selecting individual feedback resistor networks while simplifying the hardware implementation.

Another practical limitation was the unavailability of certain resistor values required for the feedback networks. In particular, the exact resistor values calculated for the **3.3 V** and **12 V** output networks were not readily available. Standard resistor values available in the local market were therefore used as substitutes or combined to obtain the closest possible resistance values. Since the output voltage of the LM2596 adjustable regulator is directly determined by the feedback resistor ratio, these substitutions resulted in deviations from the theoretical output voltages.

During experimental testing, the feedback network designed for **3.3 V** produced an output voltage of approximately **2.95 V** instead of the intended **3.3 V**. This deviation occurred because a **1.5 kΩ** resistor was used in place of the calculated **1.69 kΩ** resistor, altering the feedback ratio and consequently reducing the regulated output voltage.

The **5 V feedback network** did not achieve the expected output voltage during hardware testing. Instead, a significant voltage drop of approximately **1.5 V** was observed across the output. Several factors may have contributed to this behavior, including:

- Incorrect or non-ideal feedback resistor values due to component substitution.
- High contact resistance introduced by manual point-to-point wiring on the perf board.
- Poor solder joints or loose electrical connections causing voltage drops.
- Excessive voltage drop across the Schottky diode if an alternative diode with higher forward voltage was used.
- Incorrect feedback wiring or noise pickup caused by long feedback traces.
- Increased parasitic resistance and inductance associated with perf board construction.
- Component tolerances of locally sourced passive components.
- Possible loading effects during testing or insufficient input supply current capability.

These factors collectively may have prevented the converter from reaching the intended regulated output voltage and indicate areas requiring further investigation and hardware optimization.

The **12 V feedback network** was designed and verified successfully in the simulation environment. However, due to the unavailability of the exact **9 kΩ** feedback resistor and the limitations encountered during prototype fabrication, complete experimental validation of the 12 V output could not be achieved in the current prototype. Future iterations of the hardware will incorporate precision resistor values and a professionally fabricated PCB to fully validate this output.

Overall, the project successfully demonstrated the complete engineering workflow of designing a switching power supply from first principles, including circuit design, theoretical analysis, simulation, PCB development, hardware fabrication, manual assembly, and experimental testing. Although practical limitations in locally available components affected the hardware results, the project successfully validated the converter architecture and provided valuable insights into real-world hardware implementation, component selection, and debugging. These findings establish a strong foundation for future improvements, including the use of precision feedback resistors, optimized PCB layouts, improved switching components, and comprehensive performance validation under full-load conditions.

12. Conclusion and Future scope

12.1 Conclusion

This project successfully demonstrated the complete design, development, and experimental validation of a compact multi-output DC–DC buck converter capable of operating from a variable **16–28 V DC input** and generating regulated **3.3 V, 5 V, and 12 V** output rails. The project encompassed the entire engineering design cycle, beginning with requirement analysis, topology selection, theoretical calculations, component selection, simulation, PCB design, hardware prototyping, and experimental testing.

The converter was first designed and analyzed using **Proteus Professional**, where the switching operation, feedback control, and output regulation were successfully verified through simulation. Based on the validated design, a custom PCB was developed in **KiCad**, incorporating appropriate component placement, high-current routing, and grounding considerations for reliable converter operation.

To validate the design experimentally, a hardware prototype was assembled using a perf board with locally procured through-hole components. Due to the practical limitations of component availability, certain design modifications were made during fabrication. The originally proposed three-position rotary switch was replaced with slide switches, and standard resistor combinations were used to approximate the required feedback resistor values. Despite these practical constraints, the prototype successfully demonstrated the intended operating principle of the converter and validated the overall design methodology.

Hardware testing confirmed successful operation of the converter and provided valuable insights into the challenges associated with practical implementation, including component tolerances, manual wiring, soldering quality, and feedback network accuracy. The differences observed between the simulated and measured results highlighted the importance of precision component selection and optimized PCB implementation in switching power supply design.

Overall, this project provided extensive practical experience in power electronics, analog circuit design, switching regulators, PCB design, hardware assembly, debugging, and laboratory testing. It successfully bridged the gap between theoretical design and real-world implementation while establishing a solid foundation for future improvements and industrial-scale development.

12.2 Future scope

The following enhancements can be implemented in future iterations of the project to improve its performance, reliability, and applicability:

- **Develop a dedicated printed circuit board (PCB)** to replace the perf board, reducing parasitic effects and improving overall reliability.
- **Use precision resistor values** for the feedback network to achieve more accurate 3.3 V, 5 V, and 12 V output voltages.
- **Integrate comprehensive protection features**, including Over-Voltage Protection (OVP), Over-Current Protection (OCP), Short-Circuit Protection (SCP), Reverse Polarity Protection (RPP), and enhanced Thermal Shutdown mechanisms.
- **Optimize the PCB layout** by minimizing high-current loop areas, improving grounding techniques, and reducing electromagnetic interference (EMI).
- **Improve thermal management** through the use of optimized copper pours, thermal vias, heat sinks, and better component placement.
- **Reduce output voltage ripple** by optimizing the LC output filter and selecting lower-ESR capacitors.
- **Perform comprehensive performance characterization**, including efficiency measurements, line regulation, load regulation, transient response, and thermal analysis under different operating conditions.
- **Implement digital output selection** using a microcontroller or digital potentiometer instead of mechanical switches, enabling programmable output voltages.
- **Incorporate real-time monitoring features** such as voltage, current, temperature, and fault detection using embedded sensors and a microcontroller.
- **Expand the converter to support additional output voltages** or higher output current ratings for more demanding embedded and industrial applications.
- **Evaluate alternative switching regulator ICs** with higher switching frequencies and synchronous rectification to improve efficiency and reduce converter size.
- **Validate the design against industrial standards** such as IPC PCB design guidelines and EMI/EMC compliance requirements for enhanced reliability.

- **Design a modular power supply architecture** that can be easily integrated into embedded systems, robotics, autonomous vehicles, IoT devices, and industrial control applications.
- **Conduct long-term reliability and environmental testing**, including continuous operation, temperature cycling, and varying load conditions to assess durability under real-world operating environments.

13. Bibliography

[1] Texas Instruments, **"LM2596 SIMPLE SWITCHER® Power Converter 150-kHz 3-A Step-Down Voltage Regulator Datasheet,"** Texas Instruments Inc., Dallas, TX, USA. Available: https://www.ti.com/lit/ds/symlink/lm2596.pdf

[2] KiCad Developers, **KiCad EDA Documentation**, Available: https://docs.kicad.org

[3] Labcenter Electronics, **Proteus Professional User Guide**, Available: https://www.labcenter.com

Software Tools Used

- **Proteus Professional 8.17** – Circuit simulation and validation.
- **KiCad 8.0** – Schematic capture and PCB layout design.
- **Microsoft Word** – Technical documentation and report preparation.